

BUDGET JUSTIFICATION

NSF 26-506

PESOSE: Track 3: RedTwin: AI-Enabled Digital Twins for Continuous Red Teaming of Open-Source Ecosystems

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UCSB defines a 'year' as the fiscal year that spans from July 1 - June 30 each calendar year.

A. Senior Personnel

Support is requested for 1-month summer salary at 100% effort in year one for both Professors Vigna and Kruegel. 1-month summer salary at 50% effort is budgeted for Professors Vigna and Kruegel in year two. Professor Vigna will oversee coordination of the entire project, mentor graduate student researchers, and take responsibility for all budgets and reporting to the NSF.

All faculty salaries are at university-mandated rates and university-approved projections through the period of the award, including any applicable salary increases to cover range adjustments and merit increases. Academic salary range adjustments are projected to increase by 2% per year and merit increases by 6% every three years.

B. Other Personnel

Support is requested for two resident Graduate Student Researchers (GSR) at Step 5 during the both years of the award, at 50% during the academic year from April through June, and 100% during the summer terms from July through September.

All salaries are at university-mandated rates and university-approved projections through the period of the award, including any applicable salary increases to cover range adjustments and merit increases. GSR salaries are projected to increase 3% per year.

C. Fringe Benefits

Faculty (Summer): 10.3%

GSR: 1.2%

Composite benefit rates and projections are determined and documented at the campus level per the UC Office of Budget and Planning and have been approved by the DHHS through FY26. All rates are composite benefit rates established for UC Santa Barbara and are based upon benefits administration, Dental, Disability, Employee Support Programs, FICA Tax, Incentive Award Programs, Life Insurance, Medical, Retiree Health, retirement Benefits, Senior Management Supplement, Unemployment Insurance, Vision Benefits, and Worker's Compensation. CBR does not include accrued vacation leave, UCRP interest expense, Tuition remission, Graduate Student Health Insurance, and General, Automotive, and Employee Insurance (GAEL) where applicable.

D. Equipment

\$34,206 is requested for a server capable of supporting the research necessary to this project. Of the total \$31,746 is for the equipment itself and an additional \$2,460 will cover shipping and sales tax. A price quote reflecting the technical details of this equipment is included at the end of the budget narrative.

E. Travel

The budget includes travel for both PIs to attend one project meeting annually (\$4,717 in total costs) for the purpose of collaboration. Travel includes round-trip airfare, lodging, meals and incidental expenses. For cost-estimation purposes, the parameters of these trips are projected to be as follows:

Travel	# Persons	# Days	# Nights	Lodging	M&IE	RT Airfare
Project Meeting	2	3	2	220	92	669
Project Meeting	2	3	2	220	92	670

All estimates are based on departmental experience with similar reimbursed travel expenses and according to UC travel policy G-28 (<https://policy.ucop.edu/doc/3420365/BFB-G-28>).

G. Other Direct Costs

Other: Graduate Student Health Insurance and Tuition and Fee Remission

GSRs employed at least 35% time or at least 140 hours of a three-month quarter (all GSR appointments combined) qualify for full fee and tuition remission. This includes payment of 100% of the university's fees: Student Services Fee; Tuition, All Students; UCSB Campus Based Fees, and Health Insurance. It also includes payment of Non-Resident Tuition, Supplemental for international students, and non-resident domestic students during their first year only. Non-Resident GSRs who have advanced to candidacy are eligible for a 100% reduction in supplemental tuition for a maximum of three years after the advance to candidacy. Tuition, Fees, and Health Insurance Escalation Rates: Tuition, All Students (aka Educational Tuition) is estimated to increase by 3% per year; Non-Resident Tuition, Supplemental is estimated not to increase; Health Insurance (aka GSHIP or UCSHIP) is estimated to increase 8% each academic year.

L. Indirect Costs

The prevailing indirect cost rate of 56.5% applies. This is the DHHS negotiated, predetermined, on-campus rate for Research Projects for the period beginning 7/1/25. The rate thereafter is provisional. UCSB's indirect cost is assessed on *Modified Total Direct Costs* (MTDC). The MTDC base consists of salaries and wages, fringe benefits, materials, supplies, services, travel, and subcontracts (up to the first \$25K of each). The costs of equipment, capital expenditures, tuition remission, Grad Student Health Insurance Program (GSHIP), rental costs of off-site facilities, scholarships, fellowships, the portion of each subcontract in excess of \$25K, and the total of each MCA is excluded from the MTDC base.



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Attached is link to your configuration: <https://configurator.exxactcorp.com/configure/TS4-186830206/f6d4a510-e855-43d0-99c4-b136c9e6aab7>

Item	Quantity	Unit Price	Total Price
TS4-186830206	1	\$31,746.00	\$31,746.00
Platform	1x	- 4U Rack-mountable Server - 2x Intel Xeon Socket LGA-4677 - Intel C741 - 32x Memory Slots - 8x PCI Express X16 Slots - 0x M.2 Interfaces - 2x 10GBase-T Ethernet + 1x 1GbE Dedicated Management Port (IPMI) - 4x 3000 W (2+2) Redundant - 31.5" Depth - (No Power Cables Included)	
CPU	2x	Intel Xeon Gold 5520+ 28 Core 2.2 GHz 52.5 MB Cache Socket LGA-4677 205 W	
Memory	16x	16 GB DDR5 5600 MT/s RDIMM ECC Registered	
U.2/U.3 NVME Drive	1x	3.84 TB 2.5 U.2 NVMe PCIe 5.0 x4 SSD 15 mm	
Cables	4x	C19 to C20 Power Cable - 6FT	
Warranty	1x	3-Year Warranty on Parts and Labor with Perpetual Email and Telephone Support	
M.2 Drive	--	None	
Hard Drive / SSD	--	None	
GPU	--	None	
NVIDIA NVLink	--	None	
Network Card	--	None	
OS	--	None	
Software	--	None	
Mounting Rails	--	None	